STA2020 Tutorial 1: Simple Linear Regression

1. Below we have data comparing the actual inventory level to the computer inventory record for different items sold by a company. Both variables are measured in thousands of Rands.

Computer	Actual
2462.35	2339.25
131.25	112.55
232.15	251.45
575.35	575.65
910.05	787.95
1045.45	1029.25
862.55	916.35
1915.55	2009.25
3214.35	3451.05
1209.45	1212.25
245.75	193.15

Table 1: Regression data

A regression analysis was performed on the data and the following extract of the output obtained:

```
fit <- lm(Actual ~ Computer)
summary(fit)

##
## Call:
## lm(formula = Actual ~ Computer)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -191.36  -16.54   21.49   54.06  132.79
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept) -48.45898   44.95253  -1.078    0.309
## Computer      1.04740    0.03003  34.883 6.46e-11 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 93.76 on 9 degrees of freedom
## Multiple R-squared:  0.9927,Adjusted R-squared:  0.9918
## F-statistic:  1217 on 1 and 9 DF,  p-value: 6.46e-11
```

(a) Draw a rough scatter diagram for the data provided and comment on the relationship. Do you think running the regression analysis was appropriate?

(b) Interpret the coefficient associated with the variable `Computer`. Is there evidence of a linear association between the variables? In answering this question:

  i. State the null hypothesis

  ii. State the alternative hypothesis

  iii. State the conclusion as well as how you arrived at it.

(c) What are residuals and why are they important?

(d) State the regression assumptions.

2. A study was conducted to determine the association between age and blood pressure (bp). Regression was carried out in `R-studio` and the resultant output is shown below:

```
agebp <- read.csv("agebp.csv"); attach(agebp)
fit <- lm(bp ~ age)
summary(fit)

##
## Call:
## lm(formula = bp ~ age)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -19.354  -4.797   1.254   4.747  21.153
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)  97.0771     5.5276  17.562 2.67e-16 ***
## age           0.9493     0.1161   8.174 8.88e-09 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 9.563 on 27 degrees of freedom
## Multiple R-squared:  0.7122,Adjusted R-squared:  0.7015
## F-statistic: 66.81 on 1 and 27 DF,  p-value: 8.876e-09
```

Use the output shown to answer the following questions. Round all final answers to 2 decimal places.

(a) What is the estimated regression equation? [1]

(b) Interpret the meanings of the estimated values of $\hat{\beta}_0$ and $\hat{\beta}_1$? [3]

(c) Is there evidence of a significant linear relationship between blood presure and age? Justify your answer by referring to the appropriate values in the ouput. [3]

(d) Use the estimated regression equation to calculate the predicted blood pressure of someone who is 50 years old? [2]

(e) Calculate a 95% confidence interval for this prediction. [3]
Note that this implies $\alpha = 0.05$ and with $n = 29$, from the t tables we see that $t_{a/2,n-2} = 2.05$. Further it is given that $\bar{x} = 45.97$ and $SSx = 6779.86$. NB The interval can be calculated using

$$y \in \hat{y} \pm t_{a/2,n-2} s_{\varepsilon} \sqrt{\frac{1}{n} + \frac{(x_p - \bar{x})^2}{SSx}}$$